

A $B(c_0)$ coefficient separating left and right Asplund/tame functionals

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Source Problem

The source paper is Matan Komisarchik, *Characterizations of Asplund and Tame Functionals using Arens Products*, arXiv:2602.14317. In Definition 2.29 it defines, for a Banach algebra $\mathcal{A}$ and $\varphi \in \mathcal{A}^*$,

$$\varphi \in \text{Asp}^{(l)}(\mathcal{A}) \iff B_{\mathcal{A}} \circledast \varphi \text{ is an Asplund subset of } \mathcal{A}^*,$$

$$\varphi \in \text{Asp}^{(r)}(\mathcal{A}) \iff \varphi \circledast B_{\mathcal{A}} \text{ is an Asplund subset of } \mathcal{A}^*,$$

and analogously for left and right tame functionals. Remark 2.30 states that the relationship between left and right Asplund or tame functionals is still open.

- (1) *weakly almost periodic* if $B_{\mathcal{A}} \circledast \varphi$ is weakly relatively compact in $\mathcal{A}^*$.
- (2) *left Asplund* if $B_{\mathcal{A}} \circledast \varphi$ is an Asplund subset of $\mathcal{A}^*$.
- (3) *right Asplund* if $\varphi \circledast B_{\mathcal{A}}$ is an Asplund subset of $\mathcal{A}^*$.
- (4) *left tame* if $B_{\mathcal{A}} \circledast \varphi$ is a tame subset of $\mathcal{A}^*$.
- (5) *right tame* if $\varphi \circledast B_{\mathcal{A}}$ is a tame subset of $\mathcal{A}^*$.

We will write $wap(\mathcal{A})$, $\text{Asp}^{(l)}(\mathcal{A})$, $\text{Asp}^{(r)}(\mathcal{A})$, $\text{tame}^{(l)}(\mathcal{A})$, $\text{tame}^{(r)}(\mathcal{A})$ for the appropriate subsets of $\mathcal{A}^*$.

Remark 2.30. Note that we do not distinguish between “left” and “right” weakly almost periodic functionals since those definitions are equivalent. See [3, Cor. 1.12] for the case of semigroup algebras or Proposition 3.2 for a detailed explanation. As far as we are aware, the relationship between left and right Asplund or tame functionals is still an open question [15, Quest. 7.2].

Remark 2.31. Unlike [15], we include in these definitions all of the functionals in $\mathcal{A}^*$ rather than the left or right uniformly continuous ones. We do not use any property that rely on this restriction. This discussion is of course redundant in the case of weakly almost periodic functionals, since every such functional is uniformly continuous [29, Thm. 2].

Recall that an operator $T: V \rightarrow W$ between two Banach spaces is weakly relatively compact/Asplund/tame if $T(B_V)$ is weakly relatively compact/Asplund/tame in W , respectively. The

Answer

Theorem 1. *There is a unital Banach algebra $\mathcal{A}$ and a functional $\varphi \in \mathcal{A}^*$ such that φ is left Asplund and left tame, but φ is neither right Asplund nor right tame. Consequently the left and right notions are not equivalent. Passing to the opposite Banach algebra also gives a functional which is right Asplund and right tame but not left Asplund or left tame, so neither implication holds in general.*

Construction

Let $X = c_0$, let $\mathcal{A} = \mathcal{L}(X)$ with composition, and write e_1 for the first unit vector of c_0 . Let $\epsilon_1 = e_1^* \in X^* = \ell_1$ be the first coordinate functional. Define the matrix coefficient

$$\varphi(T) = \epsilon_1(Te_1), \quad T \in \mathcal{L}(c_0).$$

The source convention for the dual bimodule is

$$\langle S \otimes \varphi, T \rangle = \langle \varphi, TS \rangle, \quad \langle \varphi \otimes S, T \rangle = \langle \varphi, ST \rangle.$$

For $y \in X$ and $\eta \in X^*$, define

$$J_X(y)(T) = \epsilon_1(Ty), \quad J_{X^*}(\eta)(T) = \eta(Te_1).$$

Lemma 1. *The two orbit sets are exactly*

$$B_{\mathcal{A}} \otimes \varphi = J_X(B_X), \quad \varphi \otimes B_{\mathcal{A}} = J_{X^*}(B_{X^*}).$$

Proof. For $S \in B_{\mathcal{A}}$ and $T \in \mathcal{A}$,

$$(S \otimes \varphi)(T) = \varphi(TS) = \epsilon_1(T(Se_1)) = J_X(Se_1)(T).$$

Since $Se_1 \in B_X$, this proves one inclusion. Conversely, if $y \in B_X$, then

$$S_y z = \epsilon_1(z)y$$

defines an operator $S_y \in B_{\mathcal{A}}$, and $S_y e_1 = y$. Hence $J_X(y) = S_y \otimes \varphi$.

Similarly,

$$(\varphi \otimes S)(T) = \varphi(ST) = \epsilon_1(S(Te_1)) = (S^* \epsilon_1)(Te_1) = J_{X^*}(S^* \epsilon_1)(T).$$

For the reverse inclusion, given $\eta \in B_{X^*}$, define

$$R_\eta z = \eta(z)e_1.$$

Then $R_\eta \in B_{\mathcal{A}}$ and $R_\eta^* \epsilon_1 = \eta$. Thus $J_{X^*}(\eta) = \varphi \otimes R_\eta$. □

Lemma 2. *The maps $J_X : X \rightarrow \mathcal{A}^*$ and $J_{X^*} : X^* \rightarrow \mathcal{A}^*$ are isometries.*

Proof. For $y \in X$, the upper bound is immediate:

$$|J_X(y)(T)| = |\epsilon_1(Ty)| \leq \|T\| \|y\| \quad (T \in B_{\mathcal{A}}).$$

For the reverse inequality, choose $f \in B_{X^*}$ with $|f(y)| > \|y\| - \varepsilon$, and set $T_f z = f(z)e_1$. Then $T_f \in B_{\mathcal{A}}$ and

$$|J_X(y)(T_f)| = |\epsilon_1(f(y)e_1)| = |f(y)| > \|y\| - \varepsilon.$$

Letting $\varepsilon \downarrow 0$ gives $\|J_X(y)\| = \|y\|$.

For $\eta \in X^*$, again

$$|J_{X^*}(\eta)(T)| = |\eta(Te_1)| \leq \|\eta\| \|T\| \quad (T \in B_{\mathcal{A}}).$$

Choose $v \in B_X$ with $|\eta(v)| > \|\eta\| - \varepsilon$, and define $T_v z = \epsilon_1(z)v$. Then $T_v \in B_{\mathcal{A}}$, $T_v e_1 = v$, and

$$|J_{X^*}(\eta)(T_v)| = |\eta(v)| > \|\eta\| - \varepsilon.$$

Thus J_{X^*} is also an isometry. □

Asplund and Tame Status of the Two Orbits

The source paper recalls the standard facts that a Banach space V is Asplund if and only if B_V is an Asplund subset, and V is Rosenthal if and only if B_V is tame. It also uses that these subset classes are preserved by bounded linear maps; in particular they are invariant under linear isomorphism onto their closed spans.

The space c_0 is Asplund, since $c_0^* = \ell_1$ is separable. It is also Rosenthal: every subspace of c_0 has separable dual, while $\ell_1^* = \ell_\infty$ is nonseparable, so c_0 cannot contain a copy of ℓ_1 . Hence B_{c_0} is both Asplund and tame. By Lemmas 1 and 2,

$$B_{\mathcal{A}} \otimes \varphi = J_X(B_X)$$

is an isometric copy of B_{c_0} , so it is Asplund and tame. Therefore φ is left Asplund and left tame.

On the other hand $X^* = \ell_1$ is not Asplund, because it is separable but its dual ℓ_∞ is nonseparable. It is not Rosenthal, because it contains a copy of ℓ_1 , namely itself. Hence B_{ℓ_1} is neither Asplund nor tame. Since

$$\varphi \otimes B_{\mathcal{A}} = J_{X^*}(B_{X^*})$$

is an isometric copy of B_{ℓ_1} , this orbit is neither Asplund nor tame. Therefore φ is not right Asplund and not right tame.

Finally, let $\mathcal{A}^{\text{op}}$ be the opposite Banach algebra. The same Banach space and functional have left and right orbit sets interchanged in $\mathcal{A}^{\text{op}}$. Thus the opposite algebra gives the reverse separation: right Asplund and right tame need not imply left Asplund or left tame. $\square$

Verification Notes

The proof uses no numerical computation. The accompanying script `code/check_orbit_formula.py` checks the two orbit identities in finite-dimensional matrix algebras, which is only a sanity check for the algebraic formulas in Lemma 1.

The source paper notes immediately after the open-question remark that it uses all functionals in $\mathcal{A}^*$, unlike a convention in the earlier tame-functional paper where one restricts to left or right uniformly continuous functionals. This example uses the source definitions. Since $\mathcal{L}(c_0)$ is unital, the usual uniformly continuous restriction should not remove the example, but this convention-specific point is marked for human review.

Novelty Check

Before promotion, the run's cheap indexes were searched for arXiv:2602.14317, the exact source title, the phrases “left and right Asplund”, “left and right tame”, “tame functionals”, and $B(c_0)$ -based variants. No prior packet or attempt for this counterexample was found. Local source-corpus searches found the source Remark 2.30 and the later Example 5.5, which says that for $V = c_0$ and $\mathcal{A} = \mathcal{L}(V)$, all matrix coefficients are left Asplund. That source example does not identify the right orbit or give the separating counterexample above.

Web spot checks on 2026-06-30 for the exact source title, arXiv:2602.14317, “Quest. 7.2” with “tame functionals”, “left right Asplund tame functionals”, and “ $B(c_0)$ tame functionals Asplund” found the source problem and related tame-functional literature, but no exact later resolution or this $B(c_0)$ coefficient construction.

Human Review Recommendation

The recommended review focus is narrow: verify the source-paper conventions for Asplund/tame subsets and the standard invariance of these subset classes under isometric embeddings. Once that is accepted, the two orbit calculations show the separation directly.

References

- Matan Komisarchik, *Characterizations of Asplund and Tame Functionals using Arens Products*, arXiv:2602.14317, 2026.
- Matan Komisarchik, *The Banach algebra $L^1(G)$ and tame functionals*, Comment. Math. Univ. Carolin. 65(2), 131–158, 2024.
- E. Glasner and M. Megrelishvili, *Representations of dynamical systems on Banach spaces*, and related Rosenthal/tame-family results cited as Fact 2.23 in the source paper.